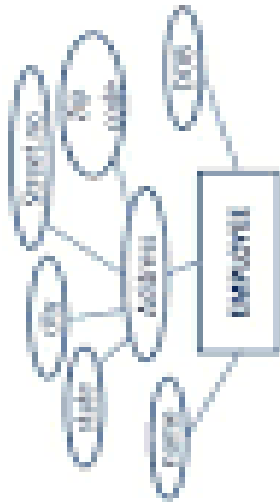


ER Model



(Entity Relationship Model)



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ER diagrams in DBMS

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Introduction To E-R Diagram



- Developed by Dr. Peter Chen in 1976, this conceptual level method, grounded in real-world perceptions, facilitates diagrammatic data representation, simplifying comprehension for non-technical users.
- An Entity-Relationship (ER) diagram is a visual representation of data within a database. It is a key tool used in database design to illustrate the logical structure of a database. This includes the various entities, their attributes, and the relationships between them. ER diagrams play a crucial role in understanding the flow of data and information within an organization's database system. They provide a clear and concise way to visualize the intertwined nature of entities and their relationships.

What is an entity in ER diagrams?

1. Definition of an Entity

An entity is a thing or an object in the real world that is distinguishable from other object based on the values of the attributes it possesses.

2. Characteristics of Entities

Entities have attributes that describe them, and they participate in relationships with other entities. Each entity in the diagram is depicted as a rectangle.

3. Types of an Entity

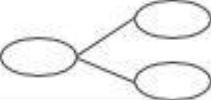
- Tangible - Entities which physically exist in real world. E.g. - Car, Pen, locker
- Intangible - Entities which exist logically. E.g. – Account, video.

4. Example of an Entity

For instance, in a university database, the "Student" entity would represent all students, while the "Course" entity would represent all the courses offered by the university.

5. Entity Set

Collection of same type of entities that share the same properties or attributes..

Entity Set 	Strong Entity Set	
	Weak Entity Set	
Attributes 	Simple Attribute	
	Composite Attribute	
	Single-valued Attribute	
	Multivalued Attribute	
	Derived Attribute	
	Null Attribute	
Relationship	Strong Relationship	
	Weak Relationship	

Attributes & its Types in ER diagrams

- Attributes are the units defines and describe properties and characteristics of entities.
- Attributes are the descriptive properties possessed by each member of an entity set. for each attribute there is a set of permitted values called domain.
- In an ER diagram attributes are represented by ellipse or oval connected to rectangle. • While in a relational model they are represented by independent column. e.g. Instructor (ID, name, salary, dept_name).
- **Example: Entity - "Student," Attributes - "StudentID," "Name," "Age"**

Types of attributes

Single-Valued Attribute

A single-valued attribute holds a single value for each entity instance. For example, "Date of Birth" for a person entity.

Multi-Valued Attribute

Multi-valued attributes can hold multiple values for an entity instance. An example is "Phone Numbers" for a person entity.

Composite Attribute

Composite attributes are made up of multiple simple attributes. For instance, an "Address" attribute may consist of "Street Name," "City," and "Zip Code" sub-attributes.

Simple

Attributes which cannot be divided further into sub parts.
E.g. Age

Relationships/Association in ER diagrams

➤ **Definition of a Relationship**

Is an association between two or more entities of same or different entity set.

➤ **Role of Relationships**

They play a vital role in structuring and organizing the data in a database, enabling the representation of complex interactions between different entities.

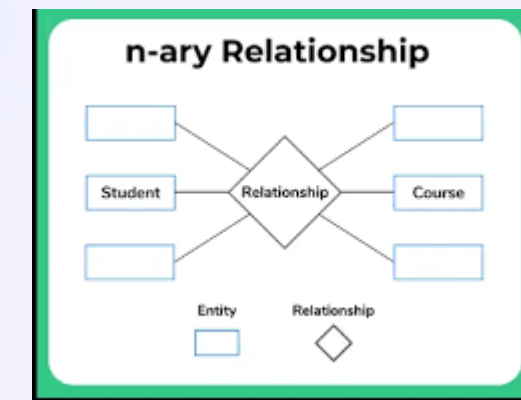
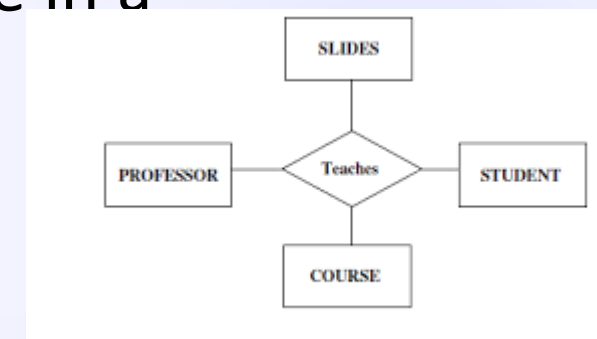
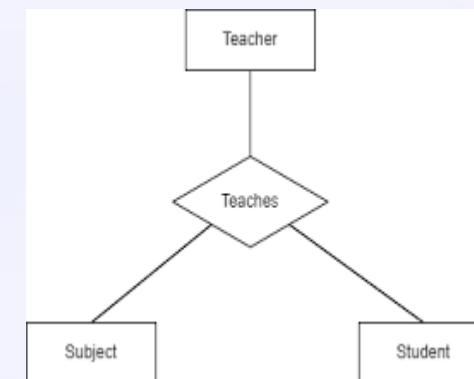
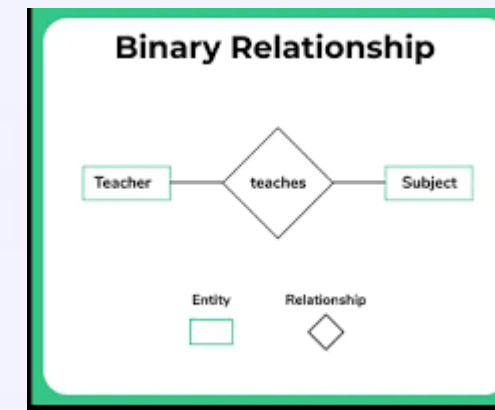
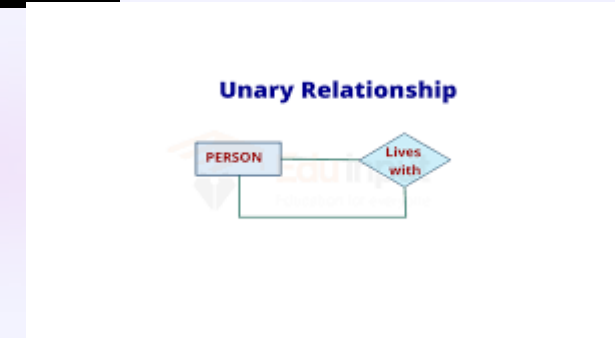
➤ In ER diagram we cannot represent individual relationship as it is an instance or data.

➤ **Every relationship type has three components.**

1. **Name** - every relation must have a unique name.
2. **Degree of Relationship** - means number of entities set(relations/tables) associated(participate) in the relationship set.
 - i. Unary Relationship.
 - ii. Binary Relationship
 - iii. Ternary Relationship
 - iv. Quaternary Relationship
 - v. N-ary relationship.
3. **Structural constraints (cardinalités ratios, participation).**

Types of relationships in ER diagrams

- **Unary Relationship** - One single entity set participate in a Relationship, means two entities of the same entity set are related to each other.
- **Binary Relationship** - Two entity sets participate in a Relationship. It is most common Relationship.
- **Ternary Relationship** - When three entities participate in a Relationship. E.g. The University might need to record which teachers taught which subjects in which courses.
- **Quaternary Relationship** - When four entities participate in a Relationship.
- **N-ary relationship** – where n number of entity set are associated.



❑ But the most common relationships in ER models are Binary.

Structural constraints (Cardinalités Ratios, Participation)

An E-R enterprise schema may define certain constraints to which the contents of a database must conform.

MAPPING CARDINALITIES / CARNINALITY RATIOS • Express the number of entities to which another entity can be associated via a relationship set.

Four possible categories are-

- One to One (1:1) Relationship.
- One to Many (1: M) Relationship.
- Many to One (M: 1) Relationship.
- Many to Many (M: N) Relationship.

One-to-One

Represents a single instance of an entity connected to a single instance of another entity.

One-to-Many

Depicts a single instance of an entity related to multiple instances of another entity.

Many-to-Many

Indicates multiple instances of an entity associated with multiple instances of another entity.

Bank Database

University Database

Inventory Management System

```
erDiagram
    Sales_Person ||--o{ Customer : "manages"
    Sales_Person ||--o{ Order : "manages"
    Customer ||--o{ Order : "places"
    Order ||--o{ OrderLine : "contains"
    Order ||--o{ Assembling : "contains"
    Product ||--o{ OrderLine : "is part of"
    Product ||--o{ Assembling : "is part of"
    Product ||--o{ Part : "is composed of"
    Part ||--o{ SupplierPart : "is supplied by"
    Supplier ||--o{ SupplierPart : "supplies"
```

The diagram illustrates the following tables and their attributes:

- Sales_Person** (green): ID : int, Name : varchar(255), Address : varchar(255)
- Customer** (yellow): ID : int, Sales_PersonID : int, Name : varchar(255), Address : varchar(255)
- Order** (blue): ID : int, CustomerID : int
- Product** (blue): ID : int, Title : varchar(255)
- OrderLine** (blue): ProductID : int, OrderID : int, Qty : int
- Assembling** (purple): ProductID : int, EmployeeID : int
- Part** (pink): ID : int, Name : varchar(100)
- Supplier** (pink): ID : int, Name : varchar(255), Address : varchar(255)
- SupplierPart** (pink): SupplierID : int, PartID : int

Relationships are indicated by lines with crow's foot notation:

- Sales_Person** to **Customer**: 1-to-many relationship.
- Sales_Person** to **Order**: 1-to-many relationship.
- Customer** to **Order**: 1-to-many relationship.
- Order** to **OrderLine**: 1-to-many relationship.
- Order** to **Assembling**: 1-to-many relationship.
- Product** to **OrderLine**: 1-to-many relationship.
- Product** to **Assembling**: 1-to-many relationship.
- Product** to **Part**: 1-to-many relationship.
- Part** to **SupplierPart**: 1-to-many relationship.
- Supplier** to **SupplierPart**: 1-to-many relationship.

Conclusion & Summary

Understanding Entities

Attributes and Types

Cardinality Ratios

Entity Sets

Relationships

Examples

ER diagrams serve as a powerful tool for visualizing and designing databases, allowing for effective communication between database designers and non-technical stakeholders. Understanding the key concepts of entities, attributes, entity sets, and relationships is essential for successful database management and design.

Thank you